# THE NUMBER OF F-EMBEDDINGS IS THE PRODUCT OF THE SEPARABLE DEGREES ALONG THE GENERATING TOWER

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The explicit counting form of the embedding bijection; the chapter uses it to compute separable degrees of finitely generated extensions.

*Editorial note.* The label lemma-count-embeddings-explicitly names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Chapter lemma-count-embeddings (delivered card L48377): Mor_F(K, overline{F}) bijects with the set of compatible root tuples (beta_1..beta_n)
- Chapter lemma-nr-roots-unchanged (source lemma): the number of roots of a polynomial is unchanged under pushforward of the coefficients by a coefficient-field homomorphism, so P_i and P_i^{phi_{i-1}} have equally many roots
- Well-definedness of the separable degree of an irreducible polynomial (observed just prior to definition-separable-degree, source lines 1480-1483): deg_s(P_i) = number of roots in any algebraic closure

*Source reference.* Original reference 09H5: sources/stacks-fields/pinned-original-source.tex, source lines 1473–1473. Prerequisite pointer invoked at 1553; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09H7: sources/stacks-fields/pinned-original-source.tex, source lines 1508–1508. Prerequisite pointer invoked at 1549; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09H6: sources/stacks-fields/pinned-original-source.tex, source lines 1486–1486. Prerequisite pointer invoked at 1544; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 09H6 we have* $|\operatorname{Mor}_F(K, \overline{F})| = \prod_{i=1}^{n} \deg_s(P_i)$.

*Proof.* This follows immediately from Lemma 09H7. Observe that a key ingredient we are tacitly using here is the well-definedness of the separable degree of an irreducible polynomial which was observed just prior to Definition 09H5. □

*Date*: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.